

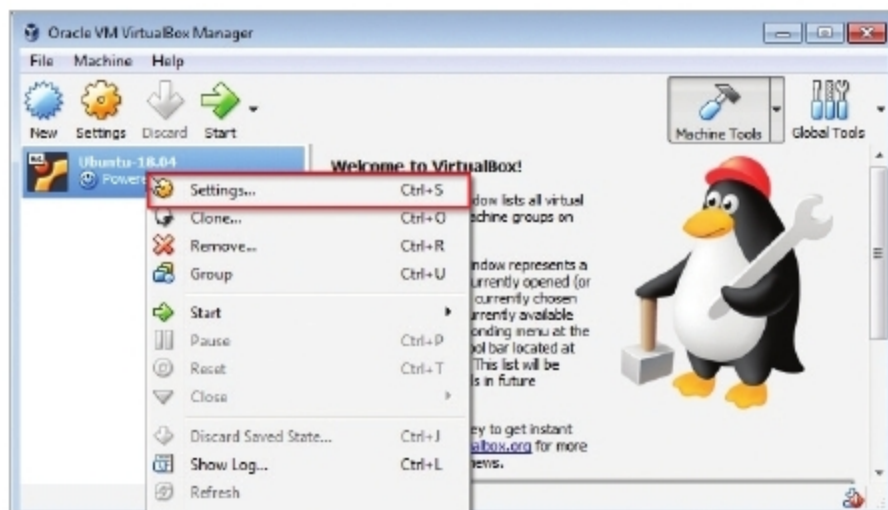


Figure 3: Configuring virtual machine

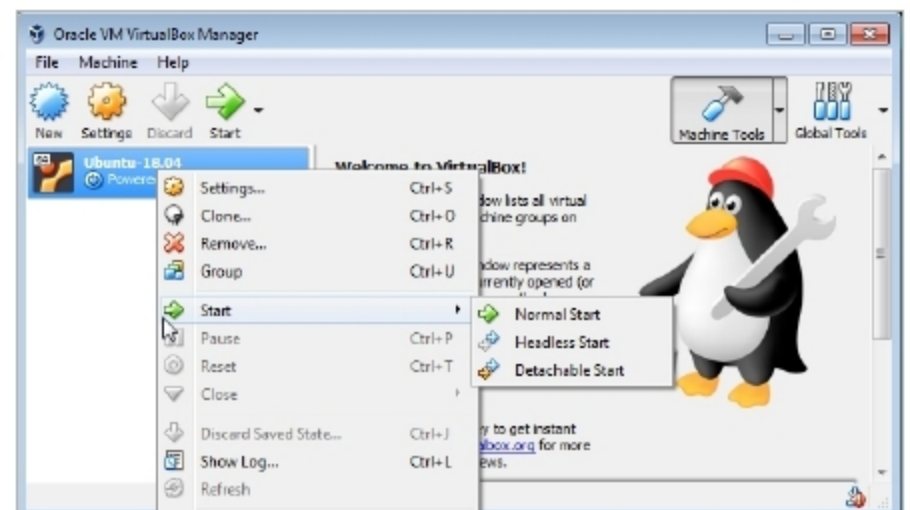


Figure 5: Starting virtual machine

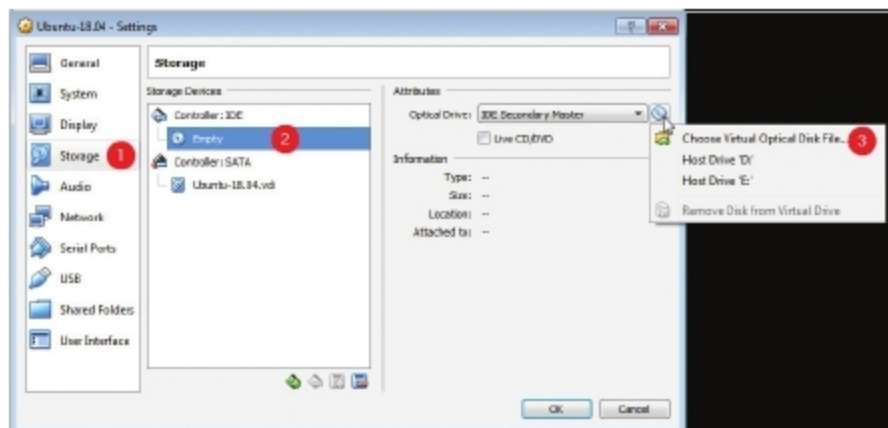


Figure 4: Selecting virtual disk

The virtual machine has now been created. This virtual machine is useless without an operating system. The next section deals with that.

Installing an operating system

To install the operating system, we need an ISO image. In this article, Ubuntu 18.04 is being used. You can download it from <https://www.ubuntu.com/download>. To install the operating system, perform the steps given below:

1. Launch the *VirtualBox manager*.
2. Right click on the newly created virtual machine and select the settings.
3. Select *Storage->Empty->Optical Drive->Choose Virtual Disk Optical File*.
4. Select the ISO image and click on the *Open* button.
5. Click on *OK*.
6. Right click on *Virtual machine* and select the *Start->Normal Start* option.
7. This will open a new window from where one can install the operating system.
8. Follow the on-screen instructions to complete the installation.

Note: To return ownership of the keyboard and mouse to your host operating system, VirtualBox reserves a special key on your keyboard for itself — the host key. By default, this is the right 'Control' key on your keyboard.

That's it, the installation is done. Now, both Windows and Linux can run simultaneously on a single computer. From now onwards, one can access the Ubuntu machine just like

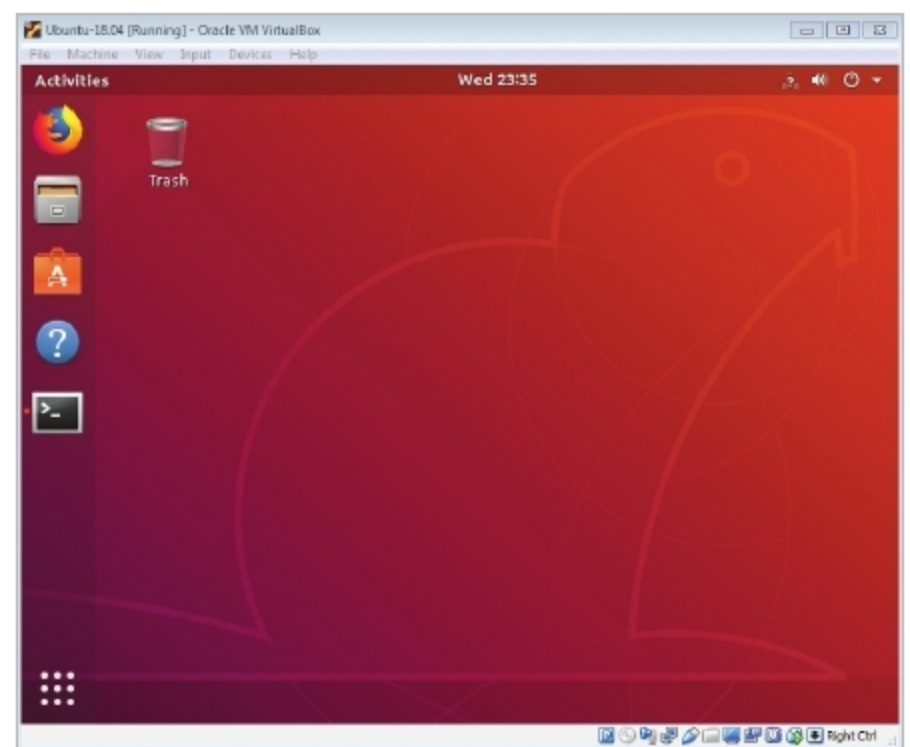


Figure 6: Ubuntu running as a virtual OS

another computer. After successful installation, you should be able to see the window shown in Figure 6.

Accessing Linux via SSH

As seen in the previous section, we can access the Linux GUI from VirtualBox itself. To access it via SSH, we need to install the SSH server. Follow the steps given below to install SSH server on Ubuntu.

1. Log in to Ubuntu from VirtualBox.
2. Open a terminal and execute the following commands:

```
$ sudo apt update
$ sudo apt install openssh-server
```

Now, let us perform the SSH login. But before that, execute the command given below by using a terminal to find the IP address:

```
$ ip addr show
```

We need a Windows SSH client for Windows and PuTTY is one of them. One can download PuTTY from <https://www.putty.org/>. To access the Linux server via ssh perform the